

Design and Implementation of a Blockchain-based Food Traceability System

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Abstract—The safety of food is intricately linked to the physical health of the general populace. However, due to the complexity of the food supply chain and the diversity of participating entities, it is often difficult to precisely identify the source of food quality issues, which in turn limits the timeliness and accuracy of liability tracing. Consequently, consumers encounter difficulties in defending their rights when faced with food quality problems. To address these issues, this paper investigates a food traceability system based on the integration of the Internet of Things (IoT) and blockchain technology. This system utilizes IoT technologies such as barcodes, QR codes, and RFID to informatively identify and track critical stages in food production, processing, storage, logistics transportation, and sales. Furthermore, by harnessing the decentralized and tamper-resistant features of blockchain technology, a secure and reliable food traceability system is established. This system not only achieves full traceability of food from the production source to the sales terminal, but also effectively ensures the transparency of food quality and safety. It possesses significant practical implications for improving food safety management, protecting consumer rights, and promoting the healthy development of the food industry.

Keywords—Blockchain; Consensus Mechanism; RFID; QR Code; Food Traceability

I. INTRODUCTION

With the continuous advancement of globalization and the increasing frequency of international trade, the food industry has become an important part of the global economy. However, at the same time, the issue of food quality and safety has gradually become prominent, attracting worldwide attention. In recent years, multiple significant food safety incidents have been continuously exposed, such as the malachite green seafood incident in China in 2005, the E. coli outbreak in the United States in 2006, the clenbuterol scandal in China in 2011, the horsemeat scandal in Europe in 2013, the palm oil milk scandal in Russia in 2015, the zombie meat incident in 2015, and the "earth pit pickled vegetables" scandal in 2022 [1]. These incidents not only pose a severe threat to consumer health, but also significantly erode the public's trust in the food industry.

Faced with increasingly severe food quality and safety issues, countries around the world have strengthened their supervision of the food industry and built traceability systems. However, traditional centralized traceability systems, due to issues such as information asymmetry, lack of trust, and data security risks, cannot meet the high standards of food safety regulation. Therefore, finding a more efficient, reliable, and transparent tracing technology has become an urgent need to address food quality and safety issues.

Blockchain technology, as a new information technology, provides new ideas for solving the problems of traditional traceability systems with its decentralized, tamper-resistant, and traceable characteristics. By building a distributed ledger, blockchain technology enables all participants to share and verify data, ensuring the transparency and authenticity of information. Meanwhile, blockchain technology utilizes encryption algorithms and consensus mechanisms to ensure the security and credibility of data. These characteristics make blockchain technology have tremendous application potential in the field of agricultural food traceability.

II. RELATED RESEARCH

Food traceability mainly focuses on recording and sharing information about all aspects of the food supply chain, including production, processing, transportation, and sales, making it accessible for regulatory authorities and consumers to query [2]. It originated in developed countries such as Europe and the United States and has gained widespread attention and application globally.

Foreign research on applying blockchain technology to the field of food traceability began earlier. Bumblauskas et al. [3] elaborated on how to utilize and apply blockchain to transport goods through global supply chains more accurately and transparently. Prashar et al. [4] proposed a blockchain-based solution that utilizes smart contracts to monitor and manage all communications and transactions among stakeholders in the supply chain network to ensure food safety. George et al. [5] introduced a traceable restaurant prototype based on blockchain and product identifiers to achieve more reliable food traceability. Kamilaris et al. [6] studied the impact of blockchain technology on agricultural and food supply chains,

pointing out that it is a promising technology for establishing an open and transparent food supply chain. Tsang et al. [7] proposed a food traceability system based on blockchain and the Internet of Things, applying new technologies such as blockchain, IoT, and fuzzy logic to a traceable food shelf-life management system, providing support for making decisions in the food supply chain. In recent years, blockchain technology has become a hot topic in China, with many enterprises, universities, and startups beginning to invest in research on this technology. Yu Zhong et al. [8] proposed a blockchain-based medical traceability system, realizing drug anti-counterfeiting and traceability information query functions. Dong Yunfeng et al. [9] applied blockchain technology to build a credible traceability model for the entire supply chain of grain, oil, and food, ensuring the security and reliability of stored data and the accuracy and credibility of traceability information.

III. RELATED RESEARCH

A. Barcode

Barcode is a graphical identifier composed of several black and white stripes of different widths according to certain coding specifications. It is an automatic information recognition technology that enables data collection, processing, and identification, and is therefore often applied in areas such as e-commerce and supply chain management. Barcodes are divided into one-dimensional codes and two-dimensional codes. One-dimensional codes are simple to produce and easy to recognize, but they can only represent simple digital information. In contrast, two-dimensional codes have a larger storage capacity, richer types of stored information, and the ability to resist damage. QR Code is currently the most widely used matrix-type two-dimensional barcode, with the following advantages: 1) It can store various types of information such as images, audio/video, fingerprints, and Chinese characters; 2) It has multiple detection points, a wide reading range, and fast speed; 3) It has anti-counterfeiting properties and strong error correction capabilities; 4) It has low production costs and is cost-effective for traceability of agricultural products such as fruits and vegetables. Based on the different characteristics of one-dimensional and two-dimensional codes, this article will adopt corresponding food information recognition technologies according to actual needs.

B. RFID

Radio Frequency Identification (RFID) is a wireless radio frequency automatic identification technology that identifies specific targets and reads relevant data through radio signals without requiring mechanical or optical contact between the identification system and the specific target. RFID systems mainly consist of three parts: tags, readers, and antennas. The tag is the data carrier of the RFID system, containing electronically stored information attached to objects to identify target objects. The reader is a device that reads tag information, emitting radio signals and receiving the tag's response. The antenna is used to transmit radio signals between the tag and the reader. In an RFID system, the reader emits radio signals of a certain frequency through the antenna. When a tag enters the reader's working area, the tag antenna generates sufficient

induced current, activating the tag and sending the information in the tag out through the antenna. After receiving the radio signal from the tag, the reader decodes the signal and sends it to the computer system for relevant data processing.

RFID technology is widely used in various industries such as logistics, retail, healthcare, and transportation. In the logistics industry, RFID technology can achieve real-time tracking and management of goods, improving logistics efficiency. In the retail industry, RFID technology can enable rapid inventory and automatic checkout of goods, enhancing the customer shopping experience. In the healthcare industry, RFID technology can be used for patient identification, drug traceability management, and other purposes to ensure medical safety. In the transportation industry, RFID technology can be used for automatic identification and management of vehicles, improving traffic management efficiency.

C. Blockchain

The concept of blockchain was initially proposed as the underlying technology of Bitcoin, a cryptocurrency. This new, decentralized, and trustless digital payment system has rapidly developed since its emergence and has caused significant impacts on traditional financial systems. Therefore, cryptocurrencies such as Bitcoin and Dash have become the initial development models of blockchain. With the advent of smart contracts, blockchain development has entered a new stage, marked by significant breakthroughs in technologies such as Ethereum and Hyperledger. The application areas of blockchain have also gradually expanded from the initial financial field to other areas such as digital evidence storage, the Internet of Vehicles, healthcare, food supply chain, smart cities, and government affairs.

Blockchain technology possesses advantages such as immutability, transparency, and traceability, which can effectively address issues in data transparency and traditional traceability systems. Data in blockchain is stored in a chain-like distributed ledger and combined with encryption technology to ensure that the data cannot be tampered with. Each node in the blockchain network performs corresponding operations based on smart contracts, and any data operations within the network will be recorded in the blockchain ledger, ensuring the authenticity, reliability, and traceability of the data. The construction of a food traceability system based on blockchain technology has the characteristics of automatic uploading of food circulation information through smart contracts and recording by all system nodes. This breaks down information silos between different nodes and enhances people's consensus on the reliability of traceability information, forming a virtuous tracing ecosystem and promoting the growth and development of the system.

IV. DESIGN AND IMPLEMENTATION OF A FOOD TRACEABILITY SYSTEM

A. Food Traceability Scheme

The system traceability scheme is illustrated in Figure 1. Food and related information are jointly stored by the blockchain and the database. The blockchain only stores the

cryptographic hash of the data, while the complete data information is stored in the latter. The database can be maintained by food regulatory authorities, while the blockchain system is collaboratively managed by all parties involved in the food supply chain.

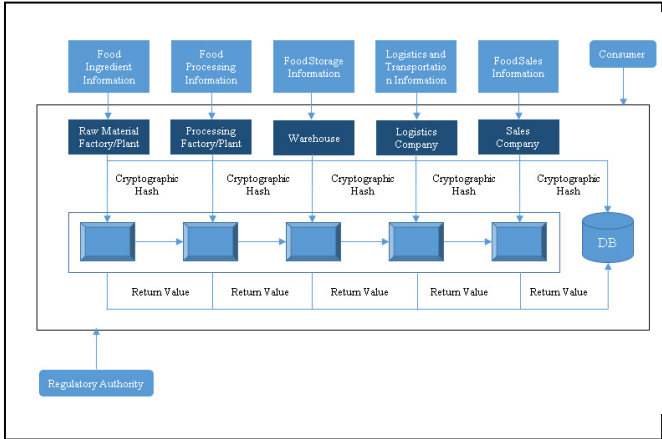


Figure 1. Food Traceability Scheme Structure Diagram

Taking a processing factory as an example, the factory collects the data information and packages it, simultaneously entering it into the factory's node data. The factory node generates a cryptographic hash of the entered data using the SHA-3 algorithm and sends it to the blockchain. After the participating nodes reach consensus, the cryptographic hash is written into a block. The block then returns a hash value that is stored in the database, which can be used as a data index. After purchasing the food, consumers can use the code on the packaging to conduct a traceability query in the database. If they are skeptical about the queried information, they can retrieve the cryptographic hash from the blockchain and compare it with the cryptographic hash generated by applying the same SHA-3 algorithm to the queried information in the database to determine whether the data has been altered. If the cryptographic hashes match, the data is authentic and unchanged. If the queried data is found to have been altered, consumers can file a complaint to the relevant departments.

B. Food Traceability Scheme

Integrating blockchain technology with existing food production management, logistics management, processing management, and sales management systems, we can establish a new blockchain-based food traceability system. This system leverages smart contract technology to automate data operations across all segments of the food supply chain, thereby reducing labor costs. Simultaneously, it enables comprehensive information monitoring and management throughout the entire process from food production to consumption, fostering a fully transparent information system that tracks food from production to shelf. The hierarchical architecture diagram of the blockchain-based food traceability system is illustrated in Figure 2.

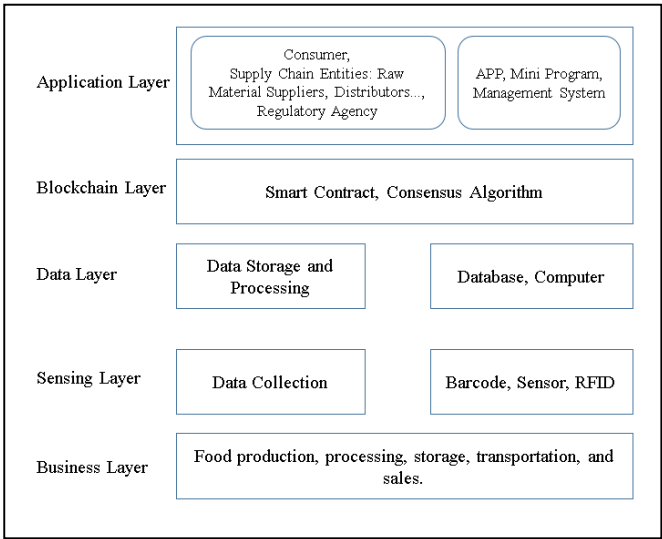


Figure 2. Food Traceability System Architecture

Operation Layer: This layer involves information collection activities essential to the food supply chain, serving as the foundation of the entire food traceability system.

Data Collection Layer: Utilizing barcode, RFID, and other information collection technologies, along with various sensors, to automatically gather and transmit data from the operation layer, enhancing the overall efficiency of the food supply chain.

Data Layer: Once food-related information is collected, it is transmitted to this layer for storage. The system employs a dual storage mechanism consisting of a relational database and blockchain. Specifically, the relational database stores complete food data along with hash values generated after digest processing, while the blockchain only stores the digest information.

Consensus & Network Layer: This layer encompasses the distributed network management, transmission mechanisms, and consensus algorithms that power the blockchain system, facilitating seamless coordination among network nodes.

Application Layer: Positioned at the top of the architecture, this layer comprises backend management systems, front-end WeChat Mini Programs, and other interfaces, primarily targeting various stakeholders in the food supply chain. Consumers, for instance, can effortlessly trace food origins through the scanning and recognition function of the WeChat Mini Program.

C. System Functional Modules

The overall design of the food traceability system based on blockchain utilizes the decentralized characteristics of blockchain to ensure the immutability of data information in the traceability process, achieving rapid search and precise location of the traceability of information flow. Based on the above overall design concept and combined with the standardized management requirements of food production enterprises, the food safety traceability system is divided into four major modules: basic information subsystem, business

management subsystem, traceability management subsystem, and traceability application program.

V. SYSTEM IMPLEMENTATION AND TESTING

A. System Implementation

This article takes the food processing module as an example to elaborate on system implementation. This module is responsible for implementing the functions of entering, modifying, and deleting key information at the processing plant node in the food supply chain. The key information includes food processing plant information, food raw material information, and processing information. This module can meet the needs of processing enterprise information management and the uploading of food processing information to the blockchain. When the food processing node enters the production information management page, the user first fills in new food processing information by clicking the upload button. After clicking to confirm and submit the form, the new data will be uploaded to the distributed database. The delete and modify buttons after each food processing information entry enable the deletion and modification of existing food processing information. The function of querying all production information operations through keywords is realized on the food information operation log page, and these food information operation logs are tamper-proof.

B. System Functionality Testing

Tests are conducted on various functional modules of the food traceability system to verify whether the system meets the design requirements in terms of business functionality implementation, further validating its usability and reliability. The testing operations are conducted from two aspects: one is reasonable data operations, and the other is unreasonable data operations. This approach is used to verify whether the traceability system can operate normally while identifying unreasonable data and preventing the execution of such operations. The test results are shown in Table 1.

Table 1. Test Results

System Testing	Items		
	Functions	Test Results	Whether It Meets the Requirements
1	User Management: Registration and Login	Successful	Yes
2	Raw Material Factory Node Information Upload and Query	Successful	Yes
3	Processing Plant Node Information Upload and Query	Successful	Yes
4	Sales Company Node Information Upload and Query	Successful	Yes
5	Consumer Node Information Upload and Query	Successful	Yes
6	Food Information Format Entry Does Not Meet Requirements	Successful	Yes

VI. CONCLUSIONS

The ability to trace food from its source and establish a reliable food distribution system is of paramount importance in ensuring food safety and protecting people's health. The food traceability system based on blockchain technology possesses superior characteristics compared to traditional systems, including its decentralized architecture, equal rights among nodes to jointly maintain system operations and ledger information, and distributed storage. By integrating blockchain technology with the Internet of Things (IoT) in the development of food traceability systems, we can first address issues such as information asymmetry, lack of timeliness, and centralized storage in current food traceability systems. Secondly, the high cost of food traceability systems has been a bottleneck hindering their development, which the blockchain-based food traceability system can effectively address. This paper utilizes various IoT technologies such as QR codes and RFID to digitize all aspects of the food supply chain. By combining these with blockchain technology, we have implemented a decentralized and tamper-proof food traceability system that well meets the modern society's demand for traceability in food safety.

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